Upper-limb rehab VR

A Meta Quest 2 app for upper-limb rehabilitation that runs the same exercises with bare hands and with controllers, and measures which input mode works better.

User Story Map

1. **User: the patient**

The user of our app is an adult doing upper-limb rehabilitation after a stroke, a fracture or hand or shoulder surgery, or, during our tests, a healthy volunteer. They sit at a table wearing a Meta Quest 2, have little or no VR experience and may have weak grip strength, so every interaction must work with slow, simple movements.

- **Goal: start a session that is adapted to their body and abilities**
  - **Activity: set up the session before exercising**
    - Start the app in a calm virtual therapy room, with a short spoken welcome and a check that the image is clear and the headset fits
    - Choose seated or standing mode and the arm to train (left, right or both)
    - Calibrate height and arm length by stretching the arm forward for 3 seconds; targets are then placed relative to this reach
  - **Activity: learn how to interact in the current mode**
    - Short tutorial for each mode: pinch and palm grab, release and poke with bare hands; grip button and trigger with controllers
    - Highlight on objects that can be grabbed and a ghost hand that shows the correct gesture
    - One practice task that is not scored
- **Goal: do the rehabilitation exercises**
  - **Activity: grasp objects and place them**
    - Objects in three shapes (ball, cube, cylinder) and three sizes appear on the table one at a time
    - The user picks up each object and places it on a marked spot; the object snaps into place when it is close enough
    - Difficulty levels change the object size, the distance to the marked spot and the number of repetitions
  - **Activity: sort objects into containers**
    - Objects in four colours and three shapes go into the container with the matching colour or shape, with the rule shown on a panel
    - A wrong placement gives soft feedback and the object returns to the table
  - **Activity: reach and touch targets**
    - Targets appear at three distances (near, medium, arm’s length), in five directions and at table or shoulder height

                - The target is touched with the index fingertip (hands) or the controller tip; it lights up and plays a sound
                - A target stays active for at most 10 seconds, then counts as missed
            - **Activity: get feedback**
                - Sound, effects and points on every successful repetition, plus a progress bar for the current exercise
                - A pause button and an automatic 30-second rest between exercises; the user can ask for a longer break
                - A warning when hand tracking is lost (hands out of view or overlapping), with a reminder to keep the hands in front of the headset
    - **Goal: try both input modes and say which one they prefer**
        - **Activity: switch to the second mode**
            - After the first block the app asks the user to pick up or put down the controllers and switches mode without restarting
            - The second block repeats the same three exercises with the same number of repetitions
        - **Activity: give their opinion**
            - SUS questionnaire (10 items, 5-point scale) after each mode, one item per screen, with large buttons pressed with a finger or the controller
            - Final question: which mode they prefer (hands, controllers or no preference)
        - **Activity: see their results**
            - Results screen comparing the two modes side by side: successful repetitions, average time, accuracy and SUS score

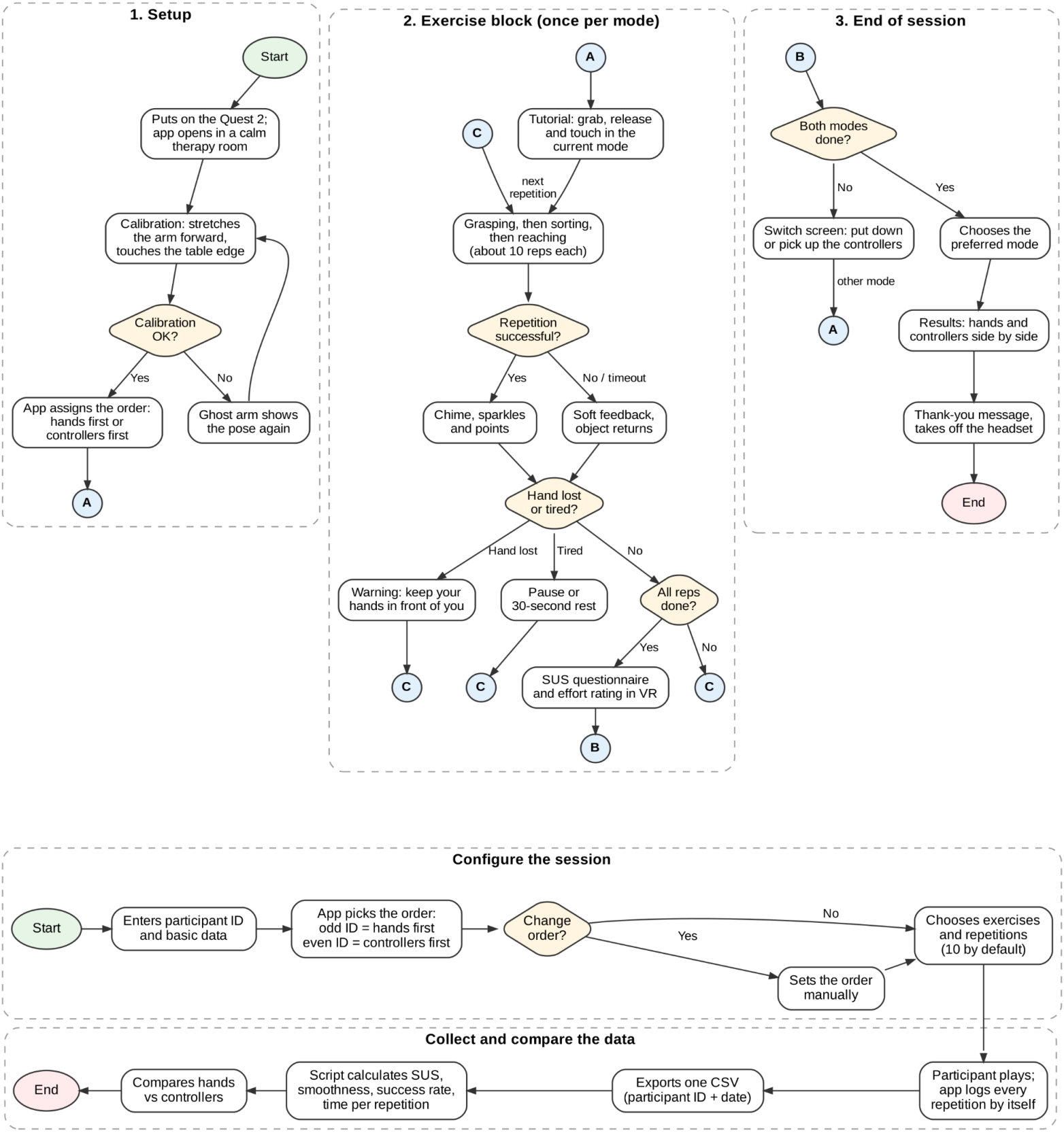
2. **User: the therapist (session operator)**

The second user is the team member or physiotherapist who runs the session. They prepare the headset, enter the participant's data, watch the session on a laptop or phone and analyse the results afterwards. Sessions must be identical across participants so the comparison is fair.

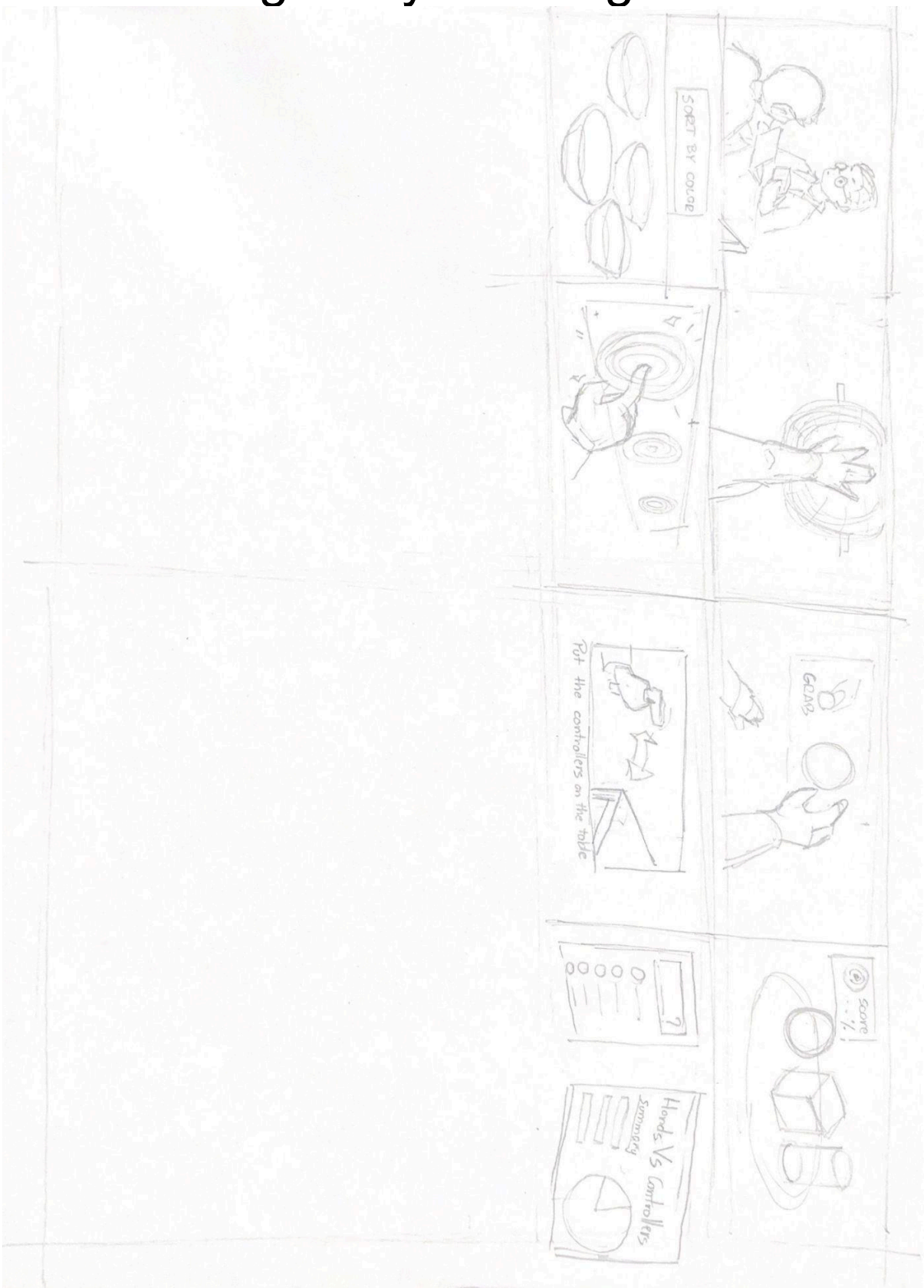
- **Goal: run a correct and comparable session**
    - **Activity: configure the session**
        - Enter the participant ID and basic data: age group, dominant hand, affected arm, previous VR experience
        - Automatic mode order: odd IDs start with hands, even IDs with controllers, with a manual override
        - Choose the exercises included and the number of repetitions for each (10 by default)

- **Goal: collect reliable data and compare the two modes**
  - **Activity: log data automatically**
    - For every repetition: participant ID, mode, exercise, target position, start and end time, success or miss, final distance to the target
    - Hand or controller position recorded every frame for trajectory and smoothness analysis
    - Tracking events: moments when hand tracking confidence is low or the hand is lost, so those repetitions can be marked
    - SUS answers, effort rating and preference saved with the session
  - **Activity: export and analyse the results**
    - One CSV file per session, with the participant ID and date in the file name
    - Script that computes SUS scores, movement smoothness, success rate and time per repetition for each mode

Journey Map (2p)



Wireframing/Storyboarding



Planning Asset Creation

3D

- ➔ Therapy room: walls, floor, ceiling and large windows (modelled in Blender)
- ➔ Exercise table
- ➔ Grasping objects: ball, cube and cylinder in three sizes (modelled in Blender)
- ➔ Sorting containers in four colours, bowl and box shapes (modelled in Blender)
- ➔ Reach targets: disc with an outer ring, idle and active states (modelled in Blender)
- ➔ Placement marker on the table for the grasping exercise (modelled in Blender)
- ➔ Decorative props (AI generated with Meshy, cleaned up in Blender: https://www.meshy.ai/)

Graphics & 2D

- ➔ Instruction panel, score and timer panel, progress bar
- ➔ SUS questionnaire screen, one item per page with five large answer buttons (own design)
- ➔ Results screen with hands and controllers side by side (own design)
- ➔ Calibration illustration: a person stretching the arm forward (AI generated, simple line style)
- ➔ Icons for hand, controller, pause, play, check, rest and settings (https://fonts.google.com/icons)

Sounds

- ➔ Calm background music, looped at low volume
- ➔ Success chime on each correct repetition
- ➔ Soft sound for a wrong container or a missed target
- ➔ Object pickup, drop on the table and drop into a container
- ➔ Short jingle at the end of each exercise and of the session (Kenney Music Jingles, CC0: https://kenney.nl/assets/music-jingles)
- ➔ Spoken instructions for the welcome, calibration, tutorial, mode switch and SUS items (generated with ElevenLabs text to speech: https://elevenlabs.io/)

VFX

- ➔ Star burst particles on every successful repetition
- ➔ Outline on grabbable objects when the hand or controller gets close (Quick Outline: https://assetstore.unity.com/packages/tools/particles-effects/quick-outline-115488)

Alte asseturi

- ➔ Meta XR All-in-One SDK (Core SDK, Interaction SDK, Building Blocks): https://assetstore.unity.com/packages/tools/integration/meta-xr-all-in-one-sdk-269657
- ➔ Interaction SDK documentation (grab, poke, hand and controller interactors): https://developers.meta.com/horizon/documentation/unity/unity-isdk-interaction-sdk-overview/
- ➔ SUS questionnaire items and scoring